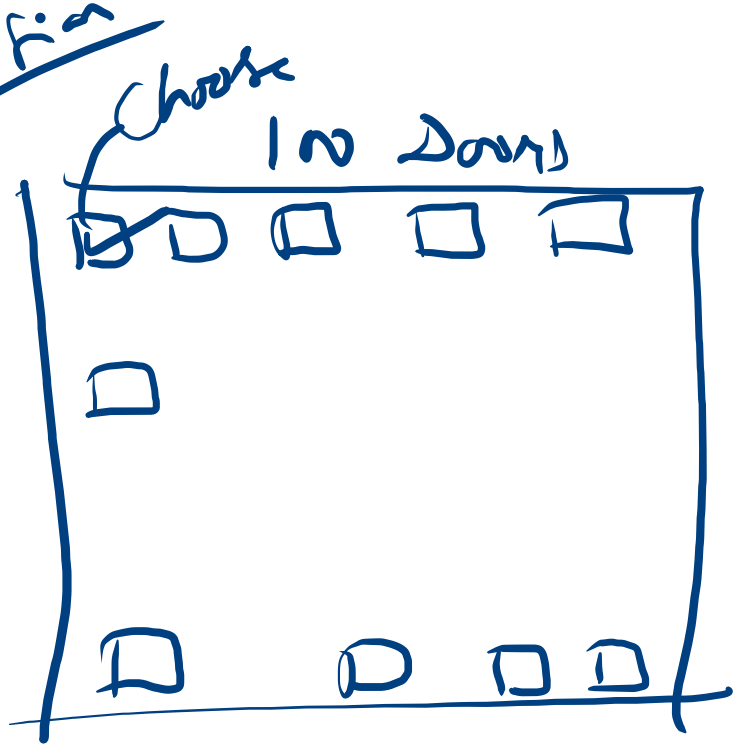
Lecture-9

Revision



Original → $\frac{1}{100}$
to each door

After all the doors (9) are shown

Random Variables

Many

RV is a variable which takes random values. X

Lets consider $x+2=7$ $\quad 2x+y=7$

x is a variable ↘ constant

x is a symbol which stands for a number.

Stand

RV is a function from the Sample Space S to Real Nos $\mathbb{R}$

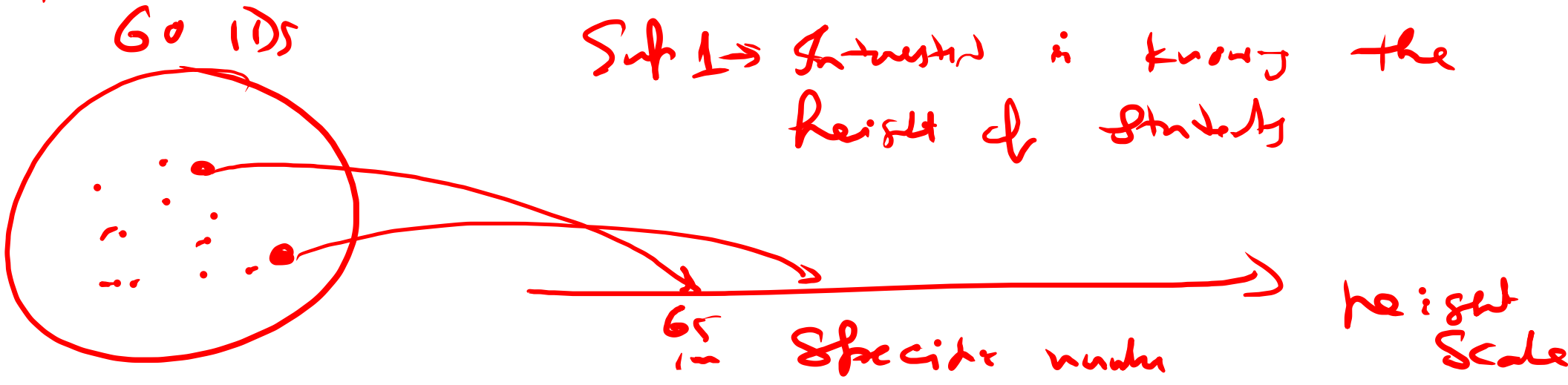


You are using DS601 exam



RV is used for associating a reward with every outcome of a random exp.

Some numerical values are more likely to occur than other numerical value → capture the probabilities of them.



Student 1 → Interested in knowing the height of student

RV → $H(\text{Student}_{ID})$ → height.

Another RV

RV → Weight W

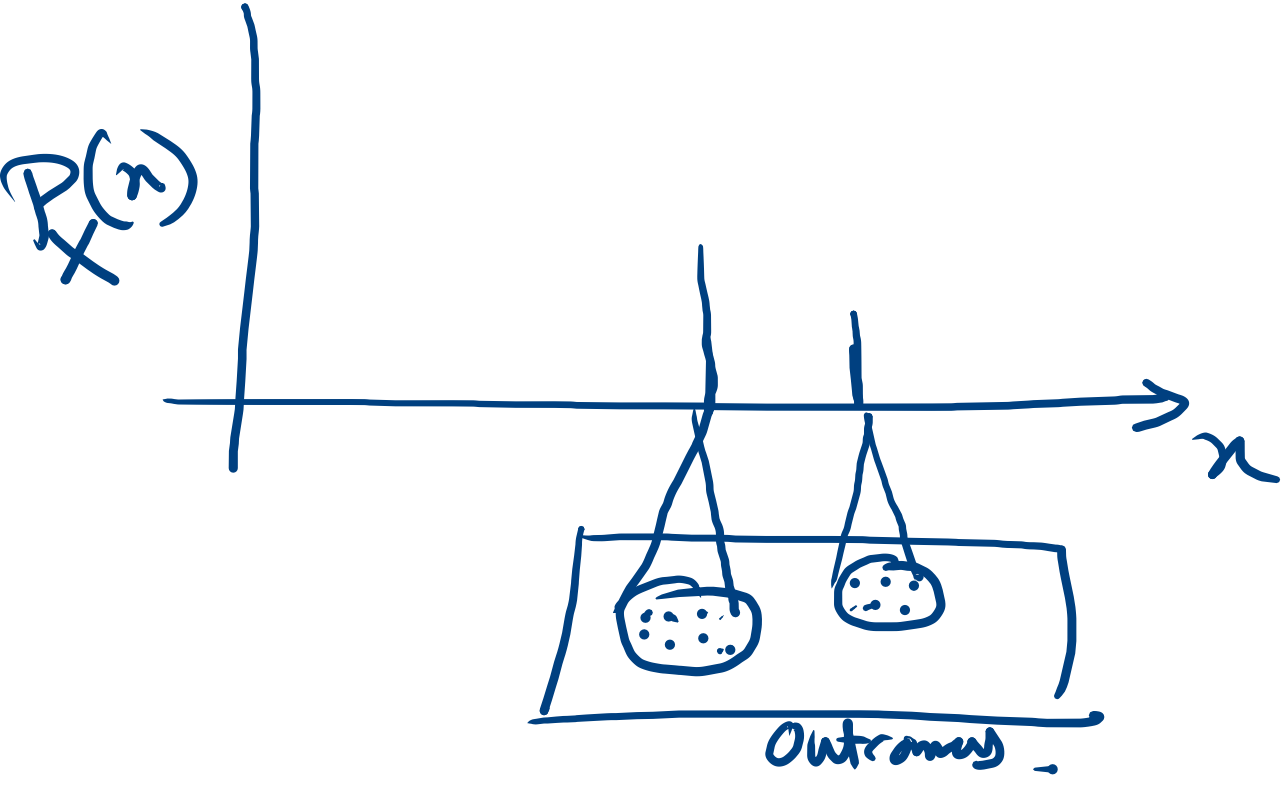
Single probabilistic event may involve several RVs.

Discrete RV

Round it off to nearest integer.

Continuous RV

Actual height upto 5 places of decimal



How likely a particular numerical value to occur.

RV X

$$P_X(x) = P(X = x)$$

Probability of RV X taking values x

$$P_X(x) \geq 0 \quad (\leq 1) \qquad \sum P_X(x) = 1$$

Ex: Start flipping a coin still I obtain a H

X → no of tosses it takes to obtain a first head.

Possible outcomes
T H
H
T T H

$X = 1$
$X = 2$
$\vdots$
$X = k$

Possible outcomes → Elements of the sample space.

Value of X depends on the outcome.

Distribution ?

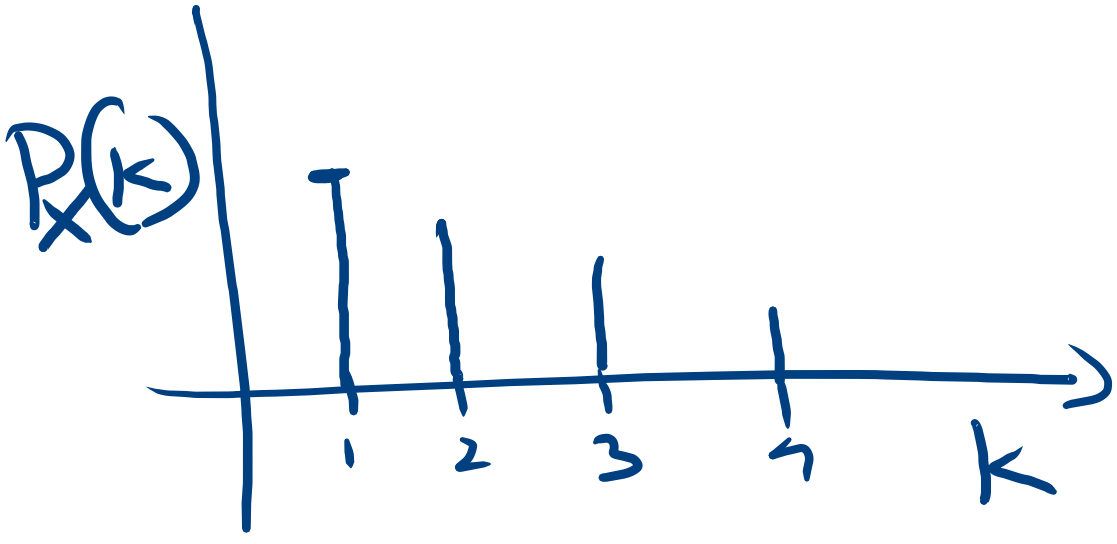
$P_X(x) = P(TT\ldots H)$

$= (1-p)^{k-1} p$

$p \rightarrow$ probability of H

$(1-p) \rightarrow$ " of T

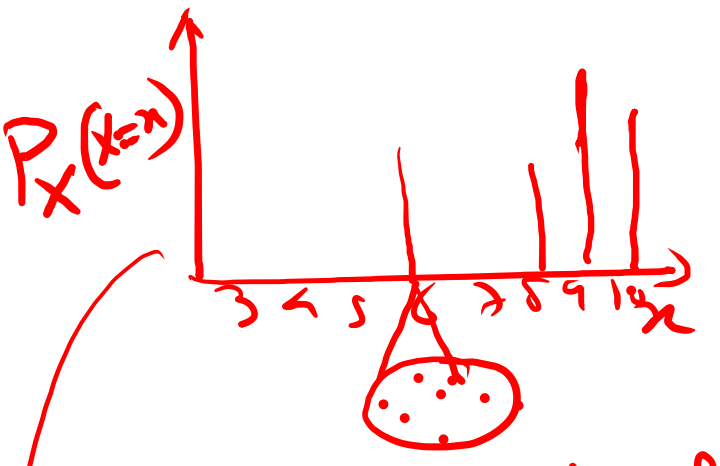
PMF →



Shape of a geometric rvs.

Lecture - 10

Keywords
RVs
Discrete & Continuous RVs
PMF



PMF

Random exp produces an uncertain outcome
RV assigns a numerical value to that outcome.

↳ how likely that we obtain an outcome that leads to that particular numerical value

N. of trials it takes to get first head

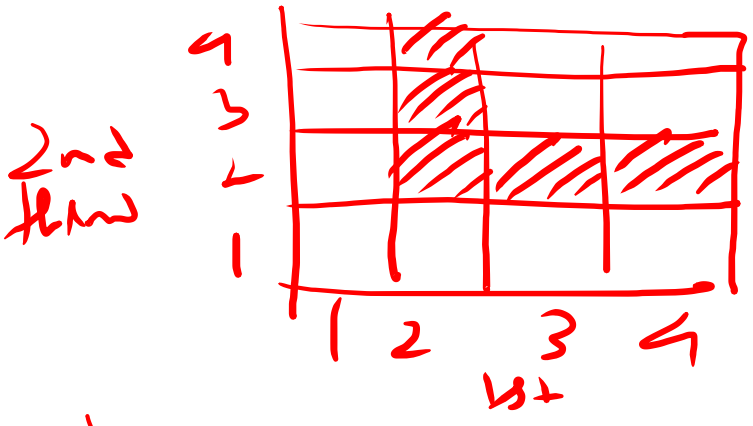
p - Success

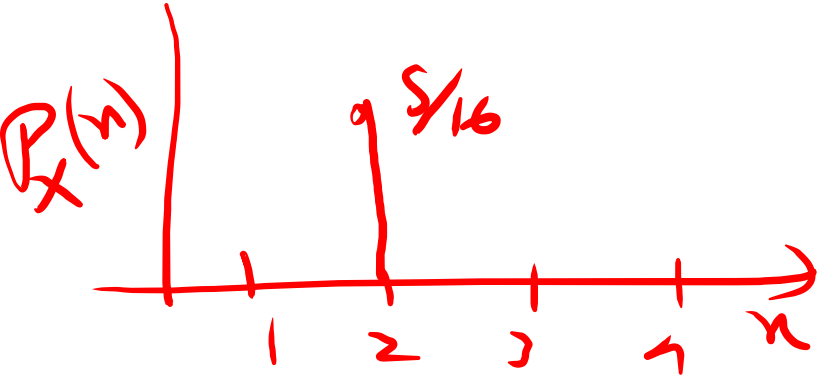


Geometric sequence

* This event can be realized in only one way.

Event: In general many outcomes can lead to same numerical value.





Two rolls of a fair tetrahedral die

First roll → outcome (1–4)

Second roll → " (1–4)

F → RV → outcome of 1st throw

S → RV → " " 2nd "

X → min(F, S) — (1–4)

$P_X(x=2)$

Def **Bernoulli distribution**

RV X is said to have a Bernoulli distribution if X has 2 possible values 0 & 1.

$$P(X=1) = p \qquad P(X=0) = 1-p$$

} Bernoulli event / trial

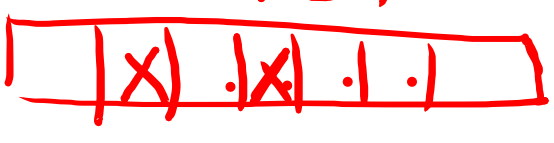
Def **Binomial distribution**

The distribution of the no of successes in n independent Bern(p) trials is called Binomial (n, p)

$$P(X=k) = {}^{n}C_{k}\, p^{k} (1-p)^{n-k}$$

k: No of successes

} 7 tosses



Binomial distribution

Realistic Experiments

Case 1 - 100% true → 10 Devices } All true

Case 2 - 96% true → 50 " } 48 true

Case 3 - 93 true → 200 " } 186 true

Which one you will choose

Your confidence is from Data points

Not one absolute %

P(Success / data) vs P(data / Success)

Models (Assumptions)

(old) [1] [2] . . . [10] [+ve] [-ve]

data available

Assume 2 more reviews

New Success rate

(New) [1] [2] . . . [5V] [+ve] [-ve] $\frac{11}{12} \rightarrow 91.7\%$

New Success rate

$\frac{47}{52} \rightarrow 94.2\%$

(New) [1] [2] . . . [20] [+ve] [-ve]

$\frac{187}{202} \rightarrow 92.6\%$

Probability of Probabilities.

What is the probability that ← 1st order P →
the long works
Probability that Success rate ← 2nd " P
is evenly 50% or 80%.

8 out of 10 80%.



Success rate

$P(+\text{ve experience}) \rightarrow .75$

Came back to the original problem

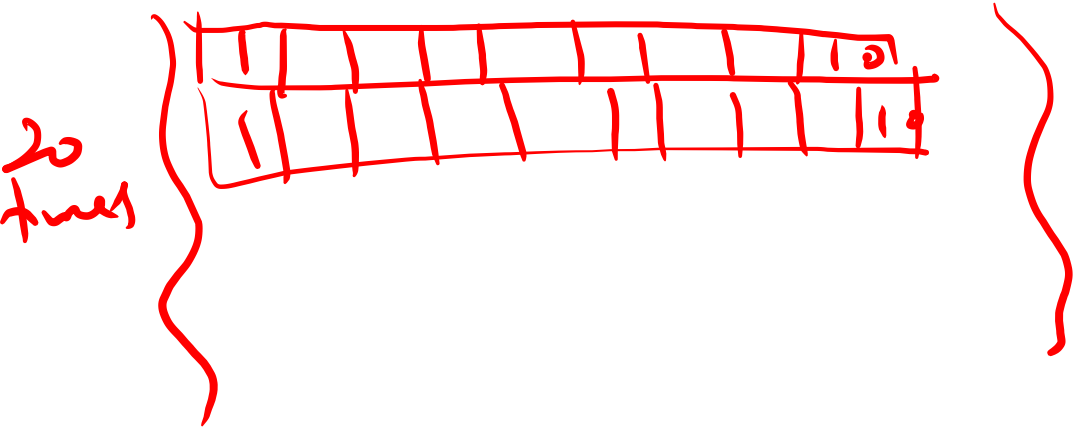
Success Rate

$\Rightarrow$ Success rate $\Rightarrow$ 100% for sure -!

it can be .95

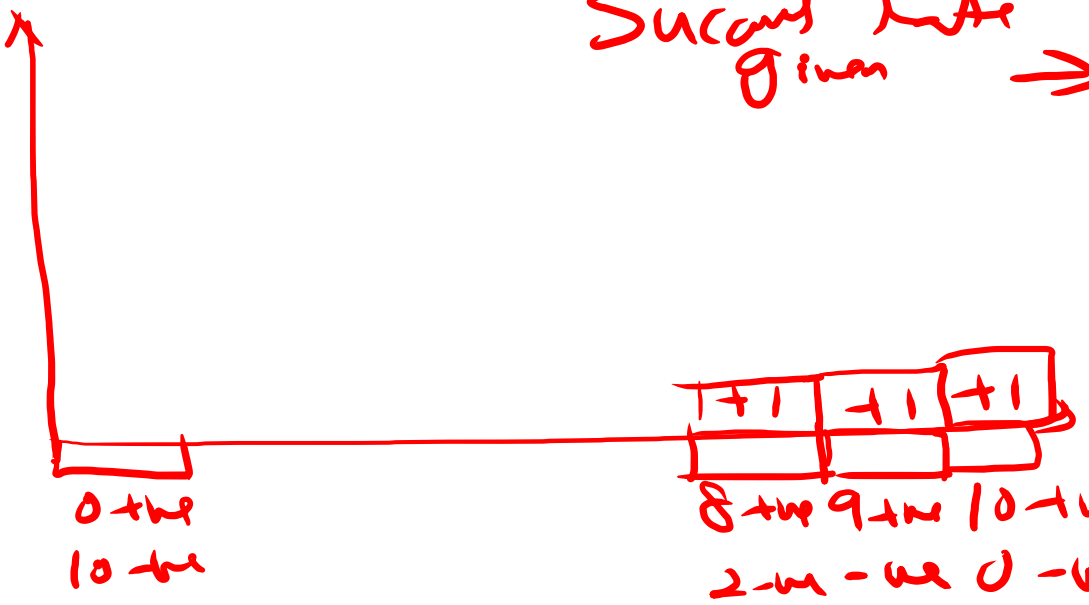
Choose a random number between [0, 1]

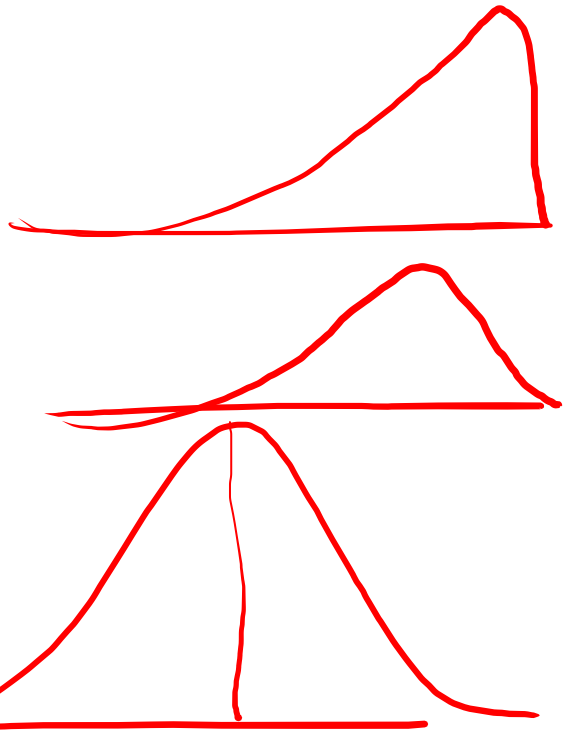
$< .95$ true $\quad > .95$ −ve



Success rate → .95

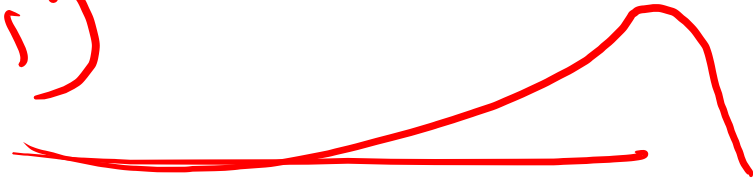
Success rate given $\rightarrow$ 95% Chance of 10





$$P(10\text{ tw}, 0\text{ ve} \mid \text{Success rate} = 95\%) = {}^{10}C_{10}(.95)^{10}(.05)^{0}$$

$$P(9\text{ tw}, 1\text{ ve} \mid \quad) = {}^{10}C_{9}(.95)^{9}(.05)^{1}$$



Lecture - 11 (Bernoulli Event)

Steps to build a distribution function (PMF)

* Define the RV
* Map the outcomes to Real numbers (Strategy)
* 

N - number of trials
p - probability of success

$$P(X=k) = {}^{n}C_{k}\, p^{k} (1-p)^{n-k}$$

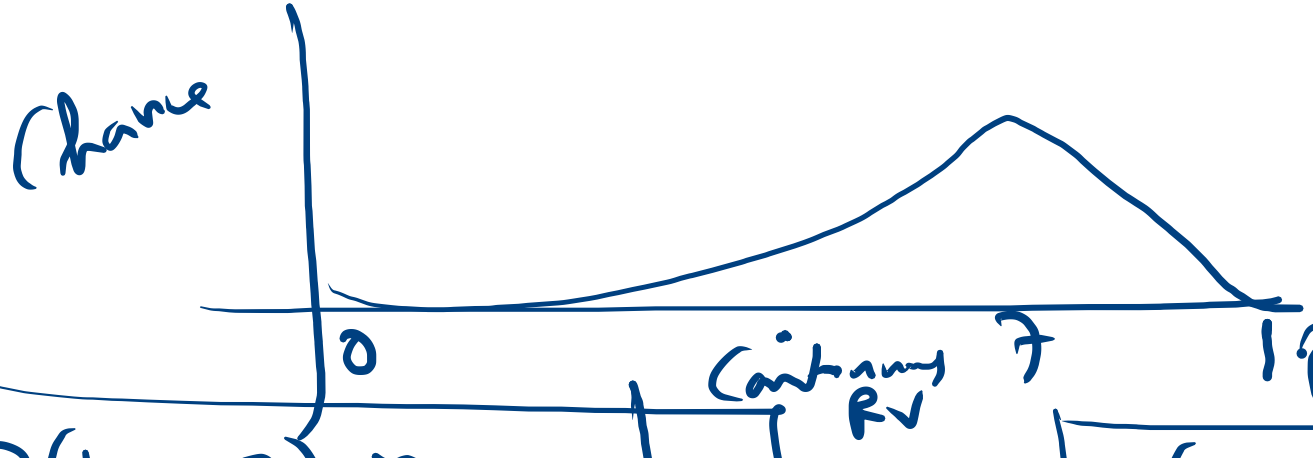
Probability of Probability

Suppose you are flipping a coin



what is $p(H) = .7$

You actually flipped 10 times and 7 times you got head



Continuous RV

Probability

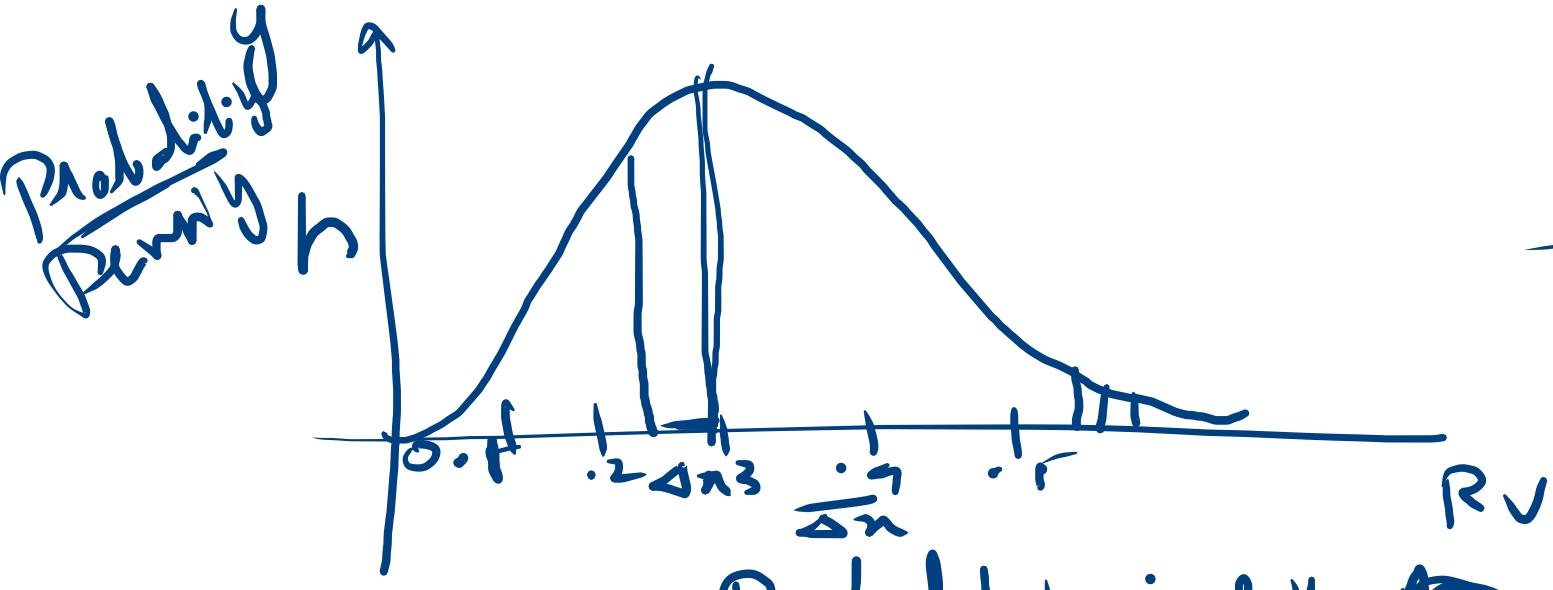
$P(P(H) = .7)$

h

$P(h = .7) \geq 0$	$= 0$
$P(h = .71) > 0$	$= 0$
$P(h = .7111) > 0$	$= 0$
$P(h = .7111111) > 0$	$= 0$

Random

Continuous RV



PDF

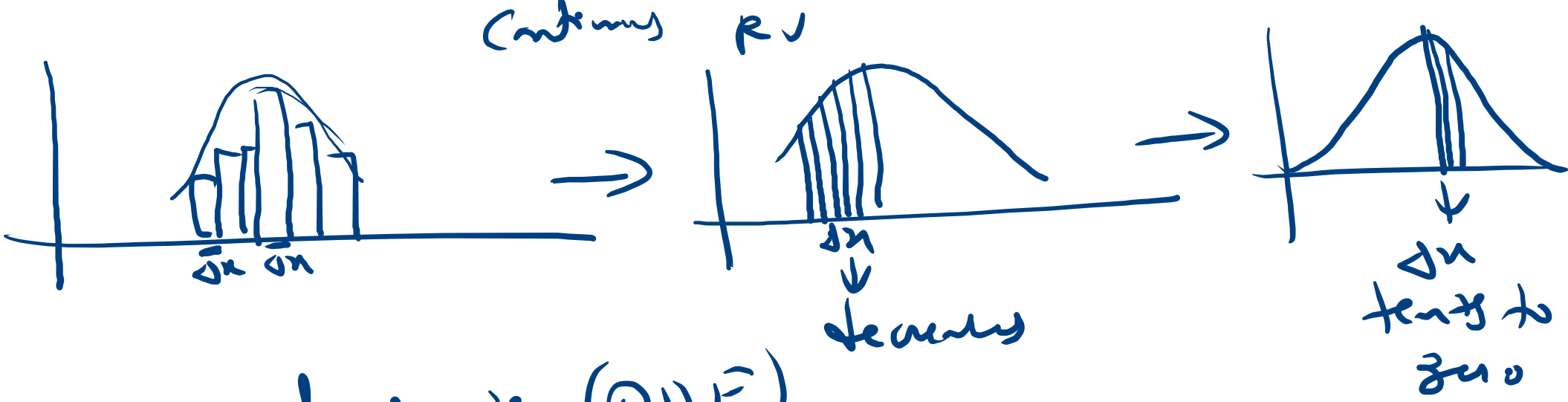
Probability is lying in the Area

$$P = h\,\Delta x$$

$$\therefore h = \frac{P}{\Delta x}$$

$$\int_0^{\infty} \text{Area} = 1$$

Continuous RV



Height represents (PDF)

$$= \frac{\text{Probability}}{\text{Per unit in the } x \text{ direction.}}$$

Smooth Curve.

Suppose you are playing a game and probability of winning Rs 1

$P(1) = \frac{1}{6}$

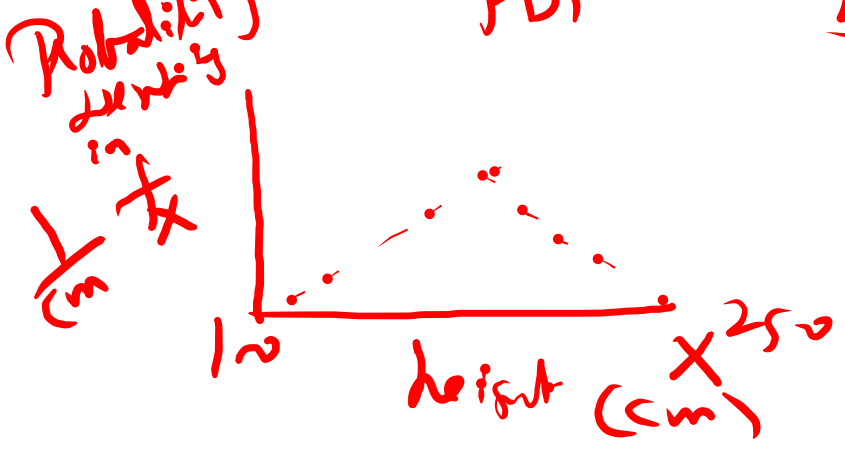
$P(2) = \frac{1}{2}$

$P(4) = \frac{1}{3}$

You want to play the game on a long run (1000 times)

PDF

Lecture - 12



$f_x(170) = .7$

Probability is finding in the area

$f_x \times \Delta x = P$

$\frac{1}{cm} \times cm = P$ (dimensionless)

Keywords
PMF
PDF

Wrong Statement - Probability of seeing exactly 170cm is 70%

Correct $P(169 \le X \le 171)$

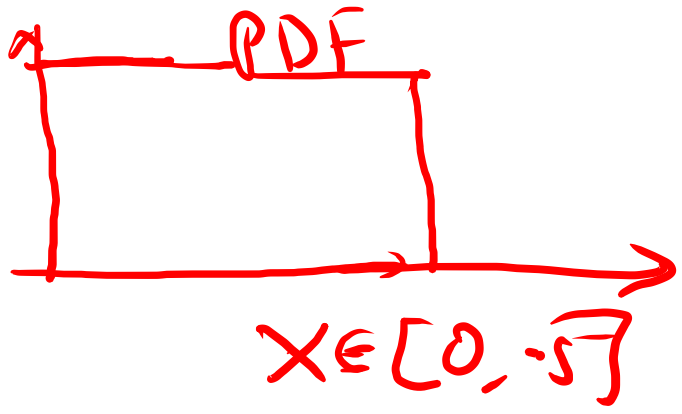
↳ around 170

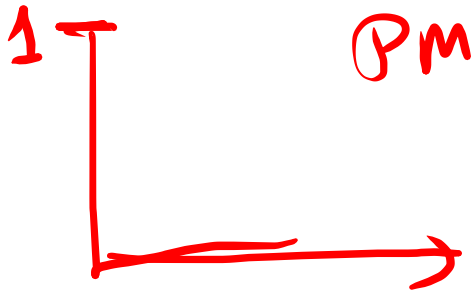
Quiz

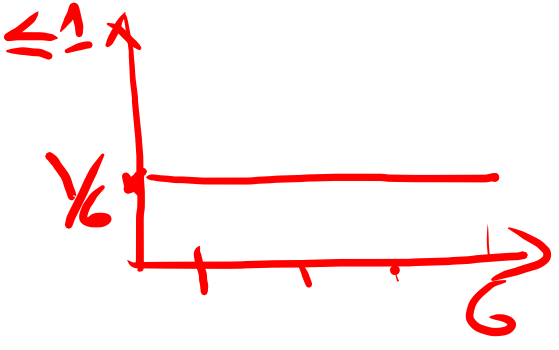
Can a PMF have Y-axis greater than 1 ? NO

" " PDF " " " " 1 . Yes

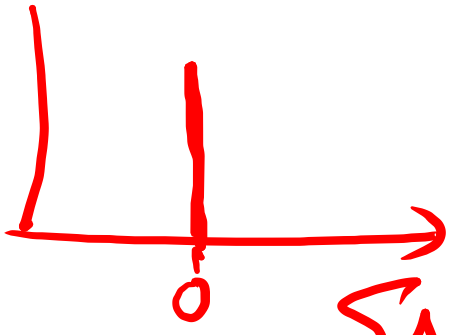
(2)?



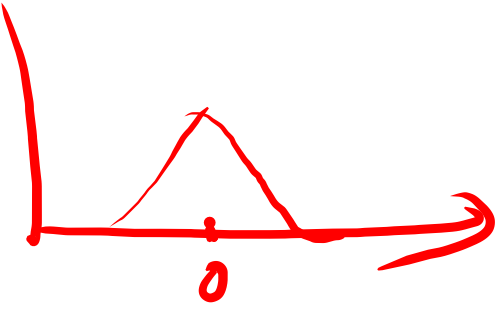


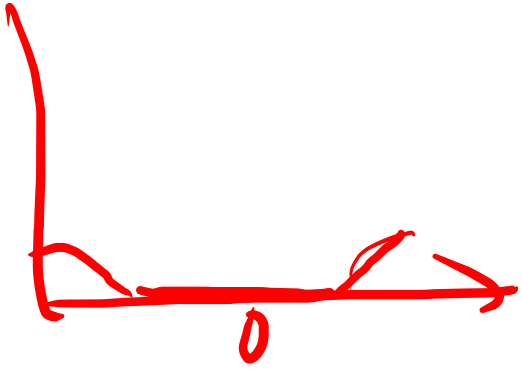


fair dice



Sinus distribution





dice

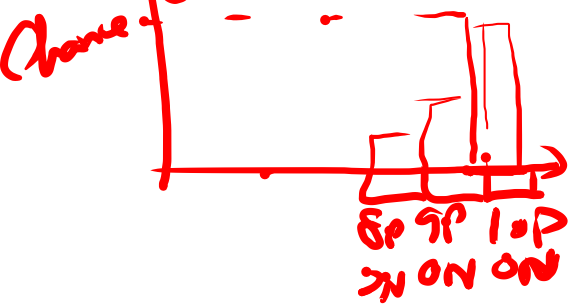


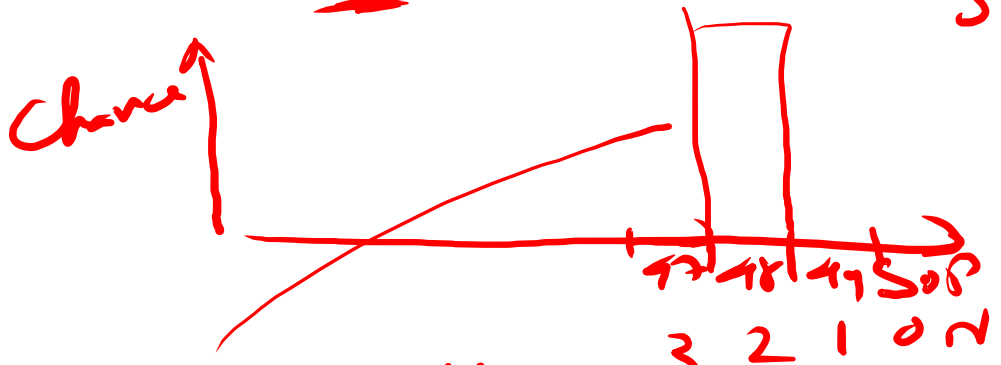
Amazon rating back (previous Lecture)

Success rate fixed = .95

Chance

60%

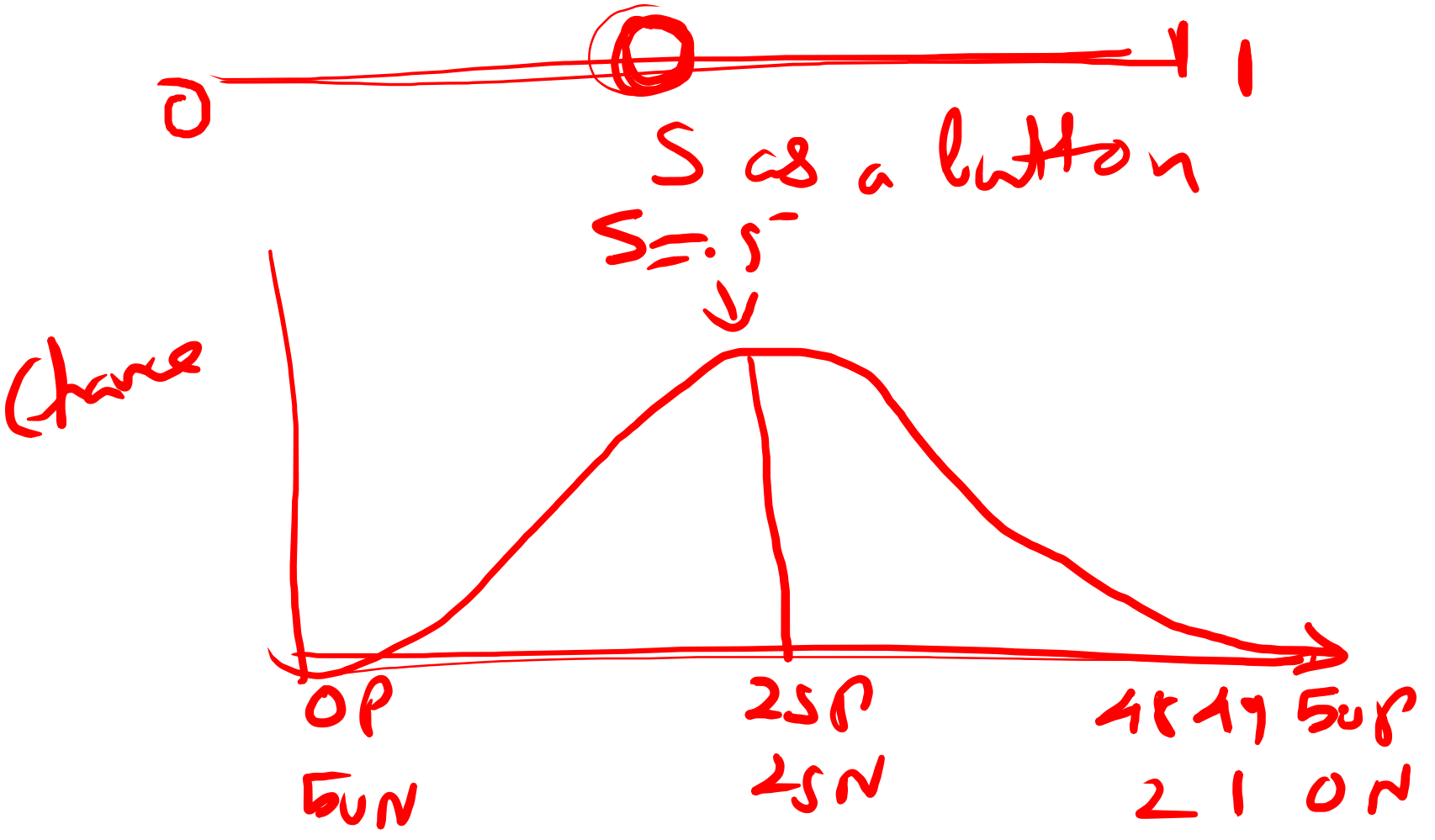




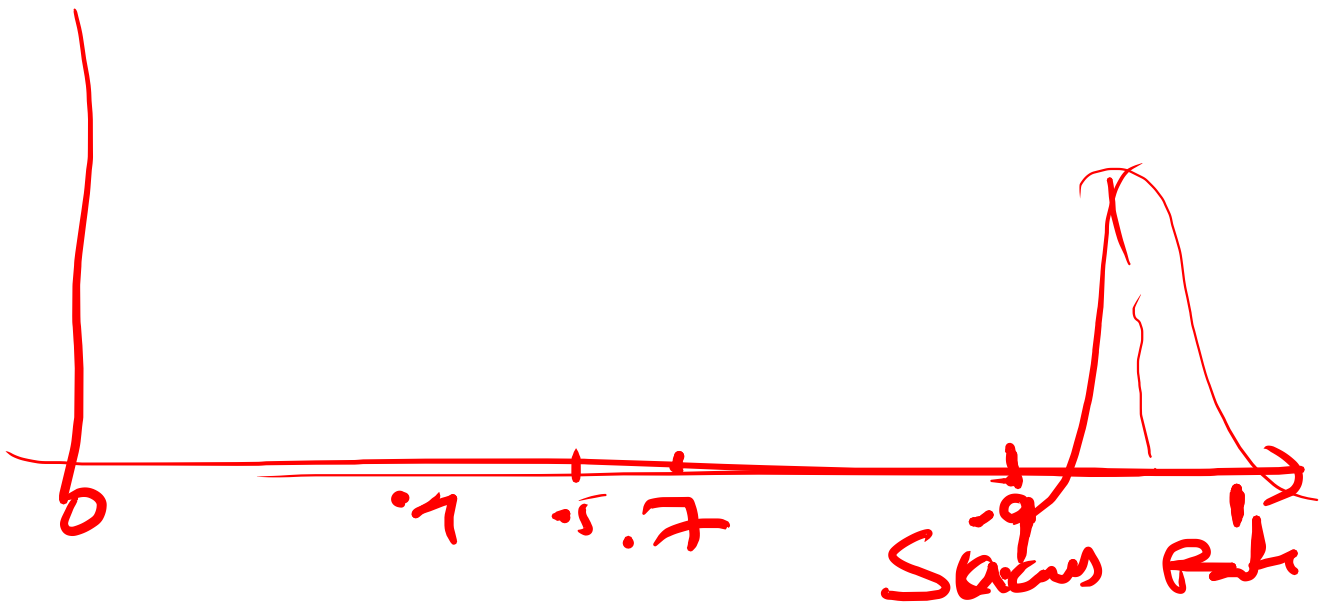
$P(48P, 2N / S = .95) = {}^{50}C_{48} (.95)^{48} (.05)^{2}$

$P(48P, 2N | S = 1) \Longrightarrow 0 \quad {}^{50}C_{48} (1)^{48} (0)^{2}$

$P(50P, 0N | S = 1) \longrightarrow 1$

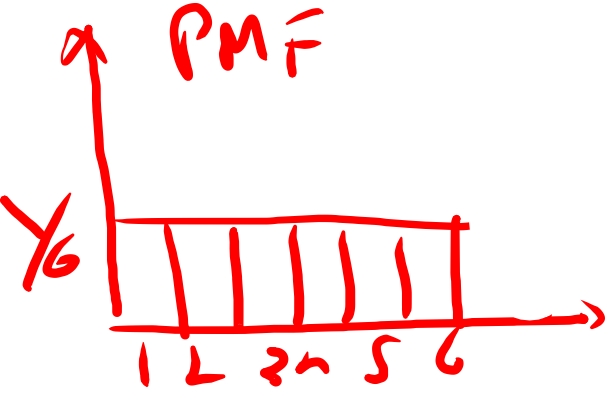

0
1
S as a button
S=.5
Chance
0P
50N
25P
25N
45 47 50P
21 0N

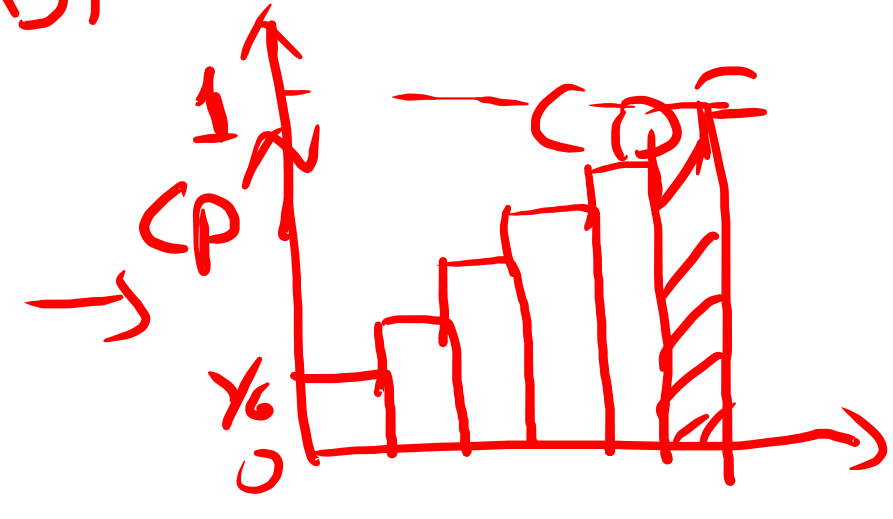
P(48P, 2N)



PMF	PDF	CDF
Y axis – Probability	Y axis – Probability Density	Y axis → Cumulative Probability

PMF
PDF > CDF

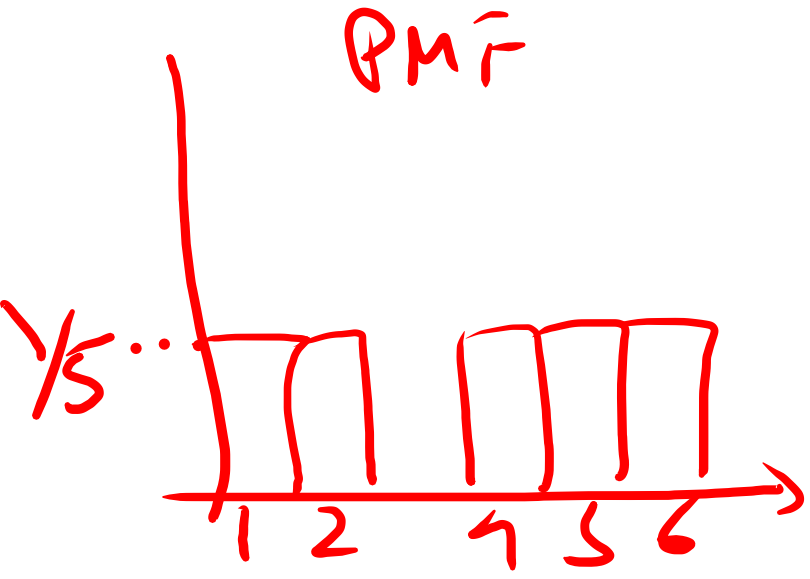


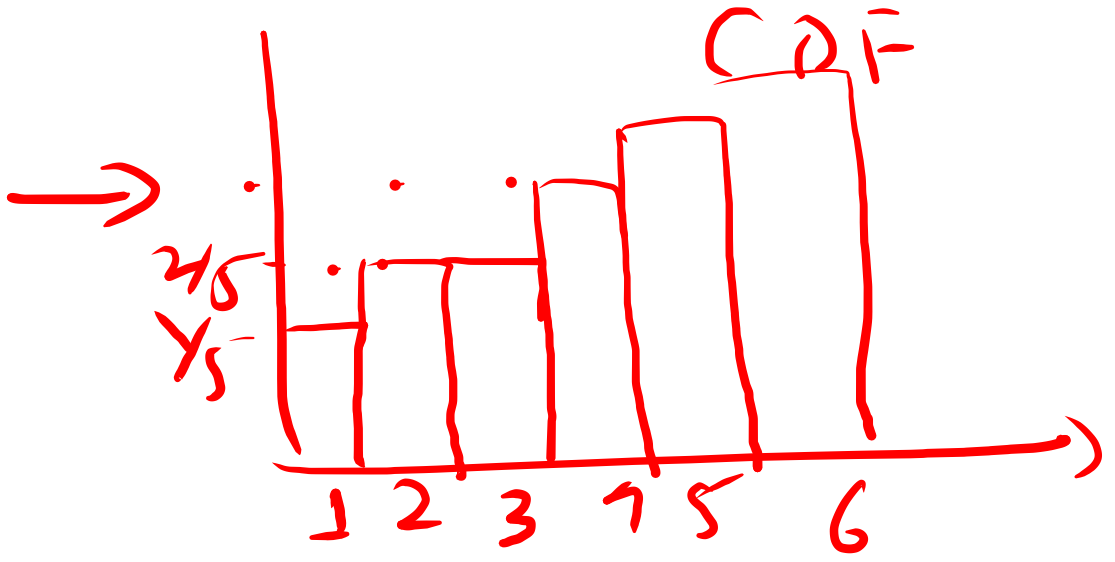




$F(X \leq 6)$

Dice which never goes 3





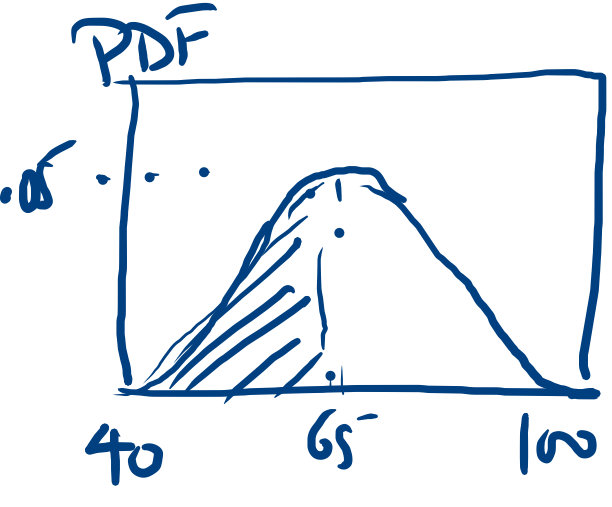
Gradient = $\frac{\Delta y}{\Delta x}$

CDF → (Gradient) → PDF → PMF, PDF

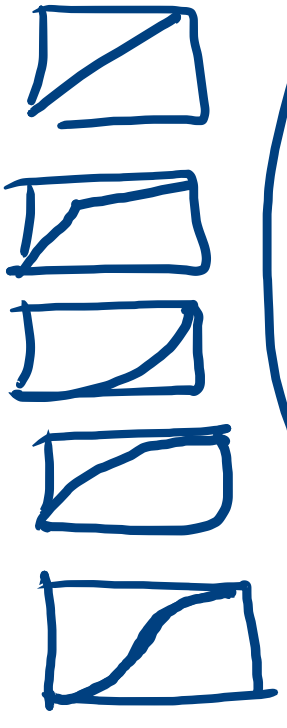
Lecture-13

CDF -> $F_X(x) = P(X \leq a)$

$0 \leq F_X(x) \leq 1$







* higher the gradient, more distribution must be concentrated around that point

Keywords

CDF

PDF

PMF

If gradient is flat, none of the distribution around that point.

PDF $\longrightarrow$ CDF

$$\int_{-\infty}^{x} f_c(x) = F_p(x)$$

CDF $\longrightarrow$ PDF

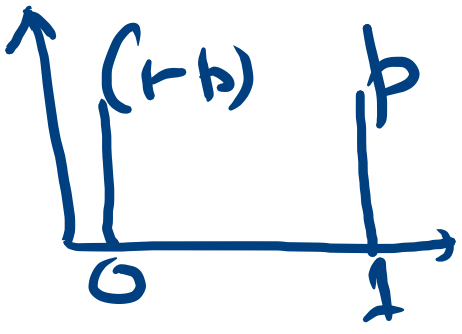
$$\frac{d\, F_x(x)}{d\, x}$$

Different kinds of RVs

1) Bernoulli with parameter $p \in [0,1]$

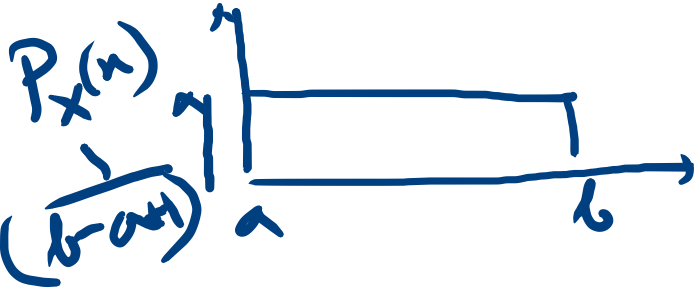
$X = \begin{cases} 1 & \text{w.p } p \\ 0 & \text{w.p } (1-p) \end{cases}$

Indicator RV



Model → Situations where trials lead to Success or failure.

2) Discrete Uniform RV



Example → Rolling a dice.

two parameters (a, b)

Range of RV

$a = 1$

$b = 6$

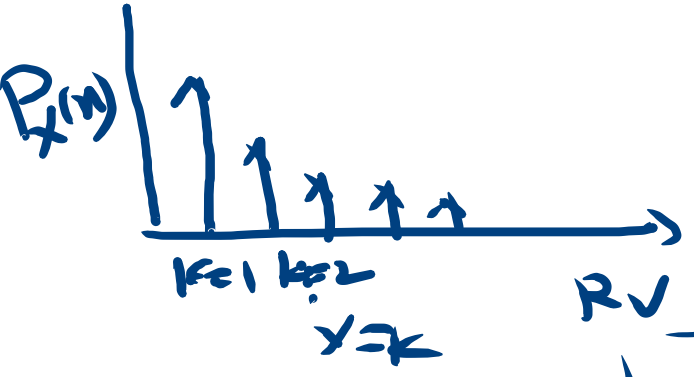
$\frac{1}{6-1} = \frac{1}{5}$

tossing a coin

3) Binomial RV Parameters n, $p \in [0-1]$

Model →

4) Geometric RV parameter ⇒ p(heads) = p



RV X: number of tosses until we get head (Success)

$$P_X(k) = (1-p)^{k-1} p$$

Problem

$P(1) = \frac{1}{4}$

$P(2) = \frac{1}{2}$

$P(4) = \frac{1}{4}$

↓

INR

You are playing a game and probabilities of winning money are given.

You play the game 1000 times. How much money on an avg you will make.

500	times	Rs 2
250	"	Rs 1
250	"	Rs 4

$$Avg = \frac{500 \times 2 + 250 \times 1 + 250 \times 4}{1000}$$

$$= \frac{500}{1000} \times 2 + \frac{250}{1000} \times 1 + \frac{250}{1000} \times 4$$

$$= \frac{1}{2} \times 2 + \frac{1}{4} \times 1 + \frac{1}{4} \times 4$$

$$E[x] = \sum_{i=1}^{n} p_x(x_i)\, x_i \quad \text{— Discrete}$$

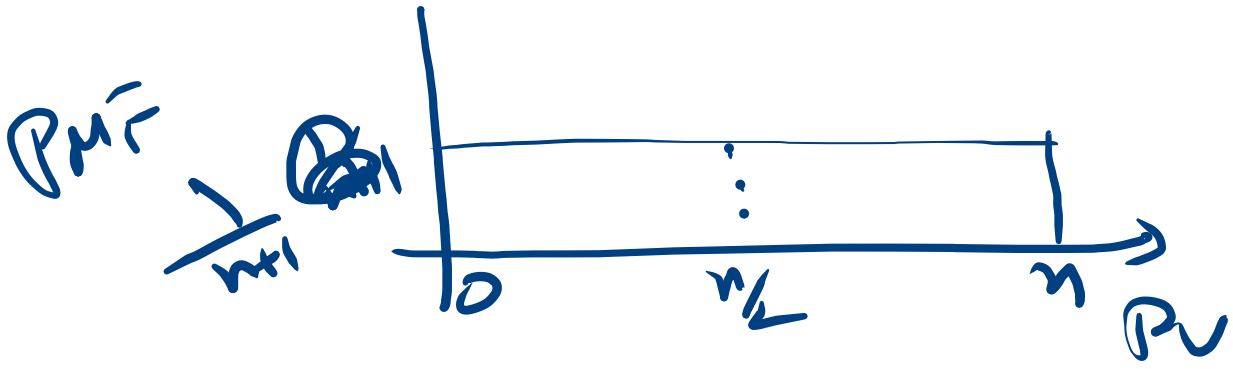
EV → Something like avg where you visualize probability as freq.

EV = probability of ith outcome × Value of the random variable associated with RV.

$$E[x] = \int_{-\infty}^{\infty} x\, f_x(x)\, dx \quad \rightarrow \text{Continuous}$$

Ex: EV of a Uniform random Variable

$(n+1) \rightarrow$ possible outcomes



$$E[x] = \frac{1}{n+1} \times 0 + \frac{1}{n+1} \times 1 + \frac{1}{n+1} 2 \cdots \frac{1}{n+1} \times n$$

$$= \frac{1}{n+1} \left[0 + 1 + \cdots n \right] = \frac{1}{n+1} \frac{n(n+1)}{2}$$

$$= \frac{n}{2}$$

You have PMF Symmetric around certain point, then point is the expected value

? $\boxed{E[X] = np}$ ✓ when X is Binomial RV n, p

Geometric RV → Expected Value.

$$E[X] = \frac{1}{p}$$

Properties of EV

1) $E[X] \geq 0$
2) $a \leq X \leq b$ then $a \leq E[X] \leq b$

3) Linearity of expectation

old salary

avg salary in a company $E[x] = a$ (avg salary)

New salary is y (2 times + 1000 Rs.)

$E[y] = 2E[x] + 1000$

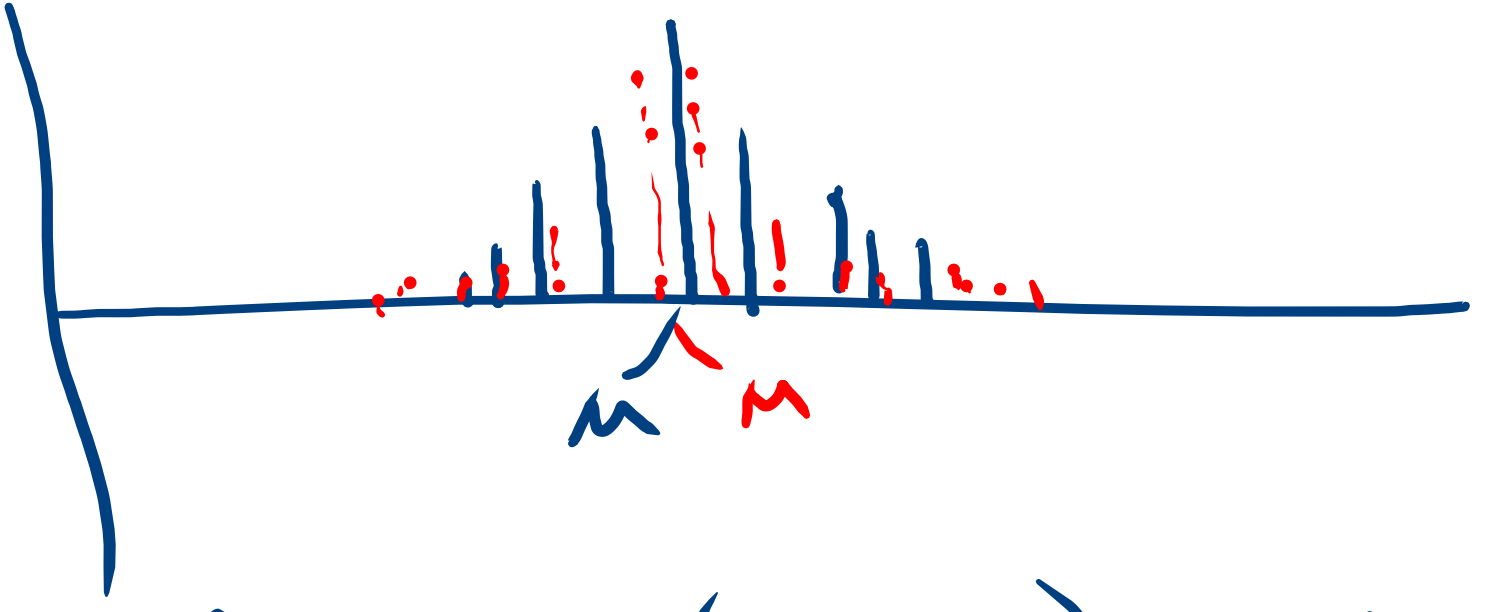
$\alpha \quad \beta$

$E[y] = \alpha E[x] + \beta$

$$E(\alpha x + \beta) = E[\alpha x] + \beta = \alpha E[x] + \beta$$

Consider a RV with given PMF

RV $\Rightarrow x$, $\mu = E[x]$



distance from mean (outcome) $\Rightarrow X - \mu$

$E[X-\mu]$

Lecture -14

$$E[x-m] = E[x] - m = m - m = 0$$

↓

mean interpretation

$$Var(x) = E\left(X - E(x)\right)^2$$

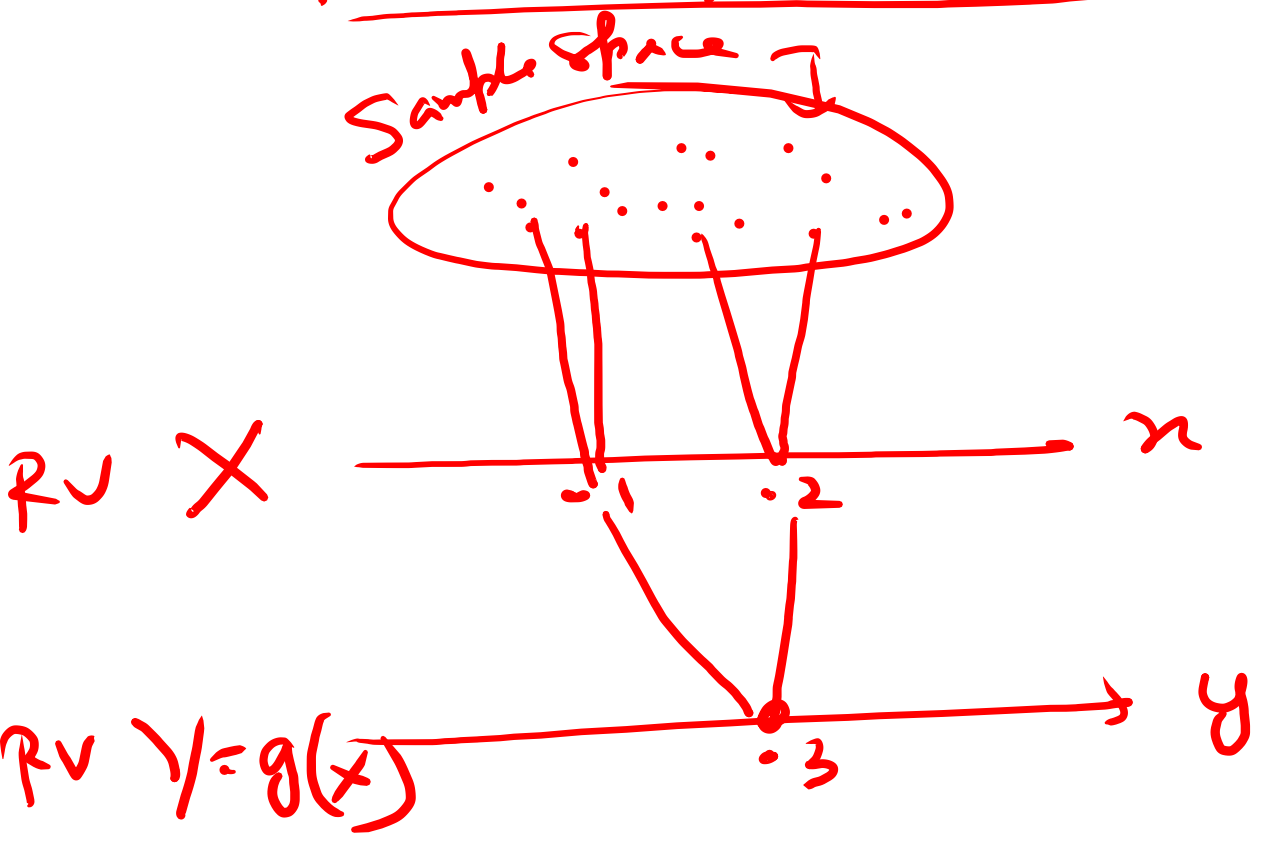
RV

$$= \sum_x (x - E[x])^2 p_x(x)$$

X - RV

E(x] → Constant

Function of a RV



$$E(Y) = \sum_y P_Y(y) \cdot y$$

$$E(Y) = \sum_x g(x)\, P_X(x)$$

$$\sum_x g(x)\, P_X(x) = \sum_y \sum_{x: g(x)=y} g(x)\, P_X(x)$$

$$= \sum_y \sum_{x: g(x)=y} y\, P_X(x) = \sum_y y \sum_{x: g(x)=y} P_X(x)$$

$$= \sum_y y\, P_Y(y)$$

$g(x)$

$$E(g(x)) = \sum_x g(x)\, P_x(x)$$

↓ function of RV

$$E(x^2) = \sum x^2 P_x(x) \neq g(E[x])$$

$$\boxed{E[x^2] \neq (E[x])^2}$$

Not true

* What happens to the Variance if we add a Constant to the RV

$Y = X + b$

RV RV Constant

Variance is unchanged

$Var(Y) = Var(X)$

$E[Y] = \nu \leftarrow$ Mean

$= \mu + b$

$\mu \Rightarrow$ Mean of X

$Var(Y) = E[(Y - \nu)^2] = E[(X + b - (\mu + b))^2]$

$= E[(X - \mu)^2] = Var(X)$

* What happens to the Var when we multiply a RV by a constant.

$$Y = aX$$

$$\boxed{Var(Y) = a^2 Var(X)}$$

$$Var(aX + b) = a^2 Var(X)$$

$$Var(5 - 4X) = 16\, Var(X)$$

$$Var(x) = E[x^2] - (E[x])^2$$

$$Var(x) = E[(x-\mu)^2]$$

$$Var(x) = E[(x-\mu)^2]$$

$$= E[x^2 - 2\mu x + \mu^2]$$

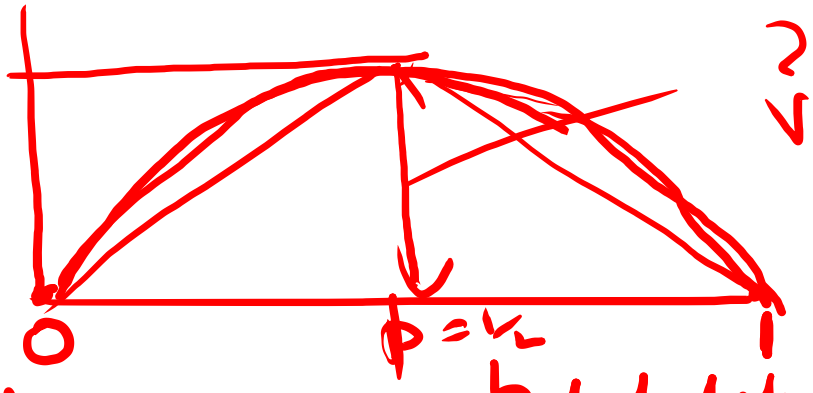
$$= E[x^2] - 2\mu E[x] + \mu^2$$

$$= E[x^2] - 2(E[x])^2 + \mu^2$$

$$= E[x^2] - (E[x])^2$$

$p(1-p)$

Variance



Value

Bernoulli Event

0 (F) → (1-p)

1 (S) → p

Bernoulli problem

$$E[x] = \sum_n P_x(n) \cdot x$$

$$= 1\,p + 0\,(1-p) = \underline{p}$$

$$Var(x) = \sum_n (x - E[x])^2 P_x(n)$$

$$= (1-p)^2 p + (0-p)^2 (1-p) = p(1-p)$$

Lecture 15

Expected value

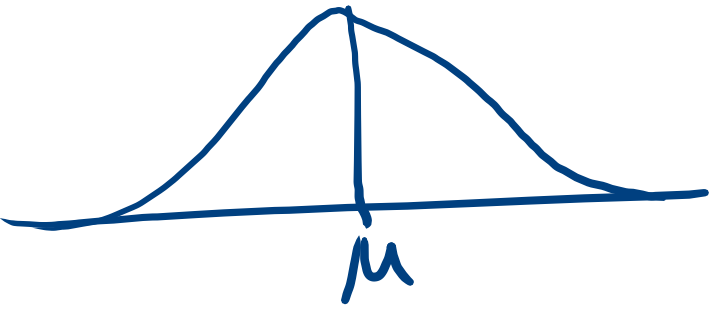
Avg, Mean

1 1 1 1 5 5 5 3 3 2

$$\text{Mean} = \frac{1+1+1+1+1+5+5+5+3+3+2}{10}$$

$$\text{Expected value} = \frac{4}{10}\times 1 + \frac{3}{10}\times 5 + \frac{2}{10}\times 3 + \frac{1}{10}\times 2$$

↑ Probability



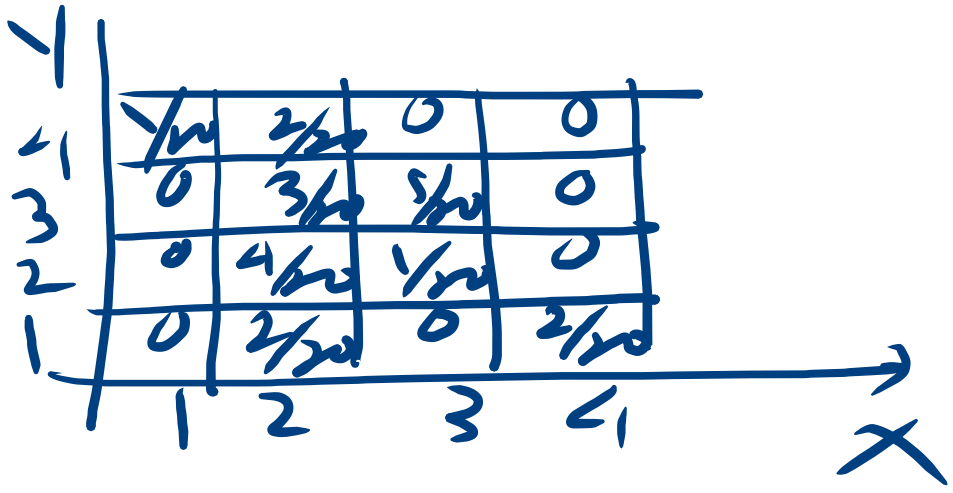
Multiple RVs & Joint PMFs

$X : p_X$
$Y : p_Y$ > Two individual PMFs

Do not give any information about how the individual RVs are related to each other.

$P(X=Y) = ?$

Joint PMF $p_{X,Y}(x,y)$ → what values of X can occur together with what values of Y.



$$P(X=1, Y=4) = \frac{1}{20}$$

$$\sum_x \sum_y P_{X,Y}(x,y) = 1$$

what is the $P_X(2) = \frac{11}{20}$

$$P_X(x) = \sum_y P_{X,Y}(x,y)$$

$$P_Y(y) = \sum_x P_{X,Y}(x,y)$$

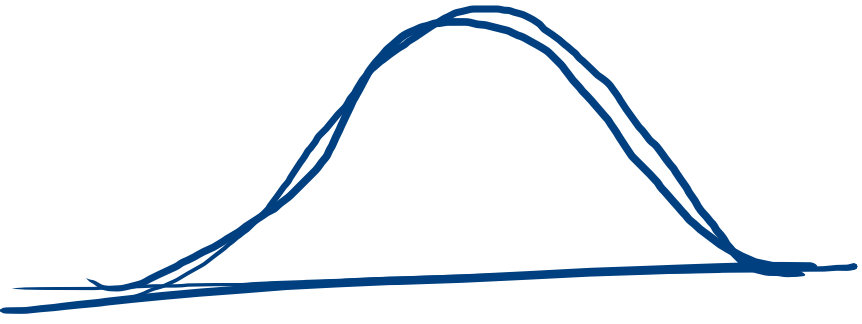
Normal RV (Gaussian RV)

Standard Normal

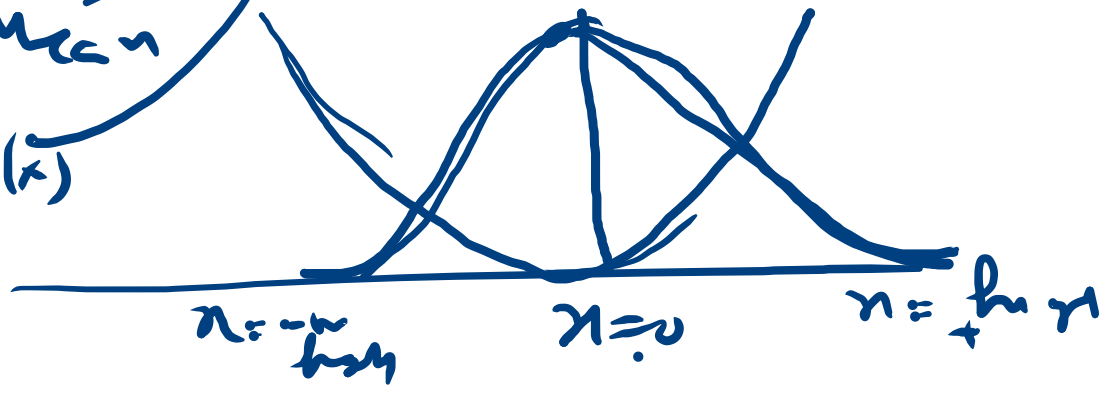
$$N(0,1) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

Mean

Var(x)



e^{-}



Calculus

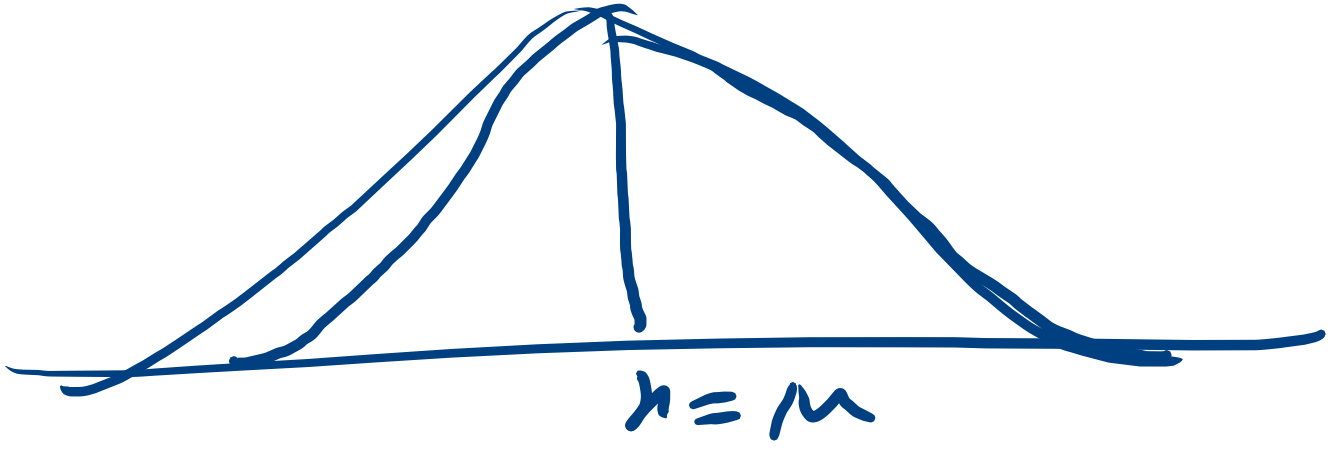
$$\int_{-\infty}^{\infty} e^{-x^2/2} = \sqrt{2\pi}$$

at $x=0, e^{-x^2/2} = 1$

$x=\infty, e^{-x^2/2} \Rightarrow 0$

General Normal RV

$$N(\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

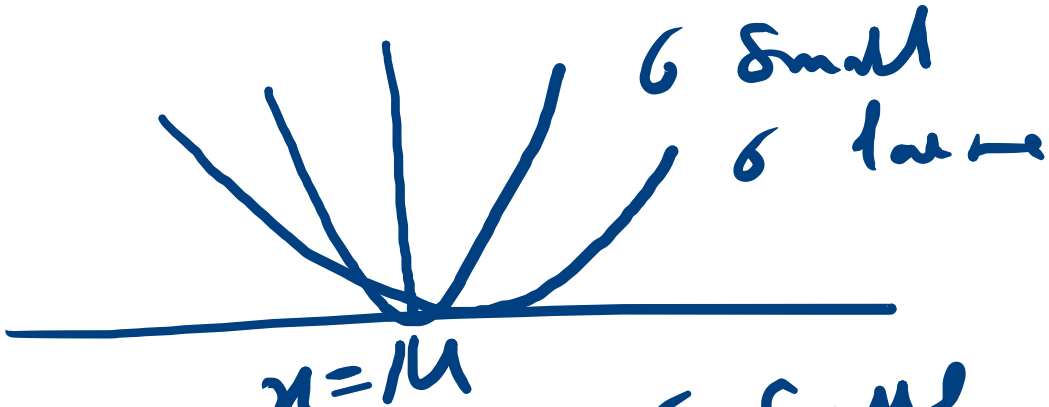


Role of σ

$\sigma \to$ Small

$\sigma \to$ large





$$\frac{(x-\mu)^2}{\sigma^2}$$



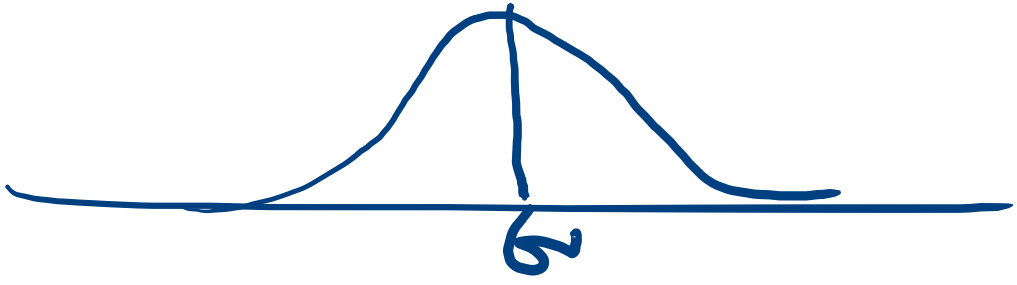
σ is small

σ is large

$$e^{-\frac{(x-\mu)^2}{\sigma^2}}$$

Why normal distribution is so common

(1) About the mean

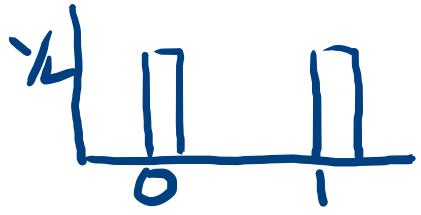


1	[illegible]
2	5 [illegible]
3	6 1
4	5 9

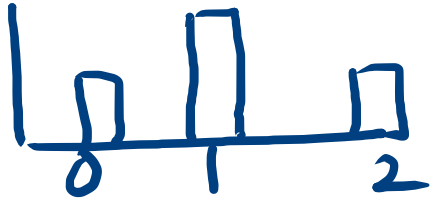
flip a Coin

No. of flips

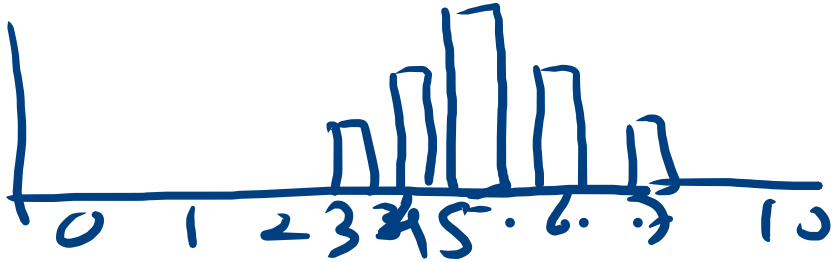
① getting a H



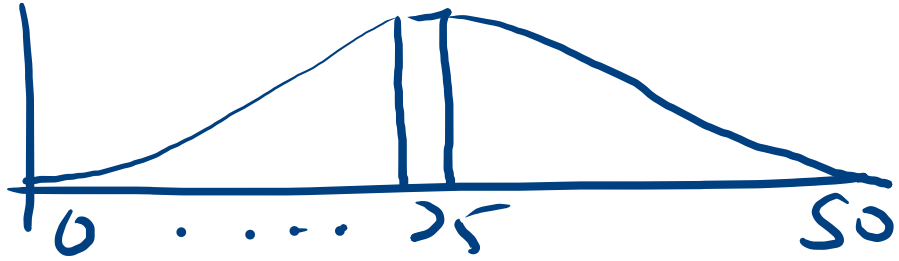
②



⑩



50 flips



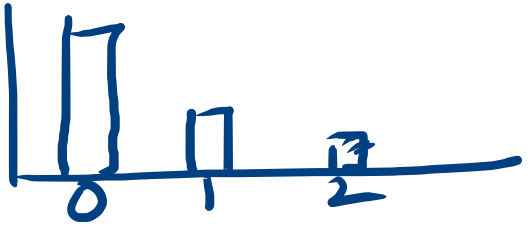
Rolling/Rolling a dice

Rolling a Sin

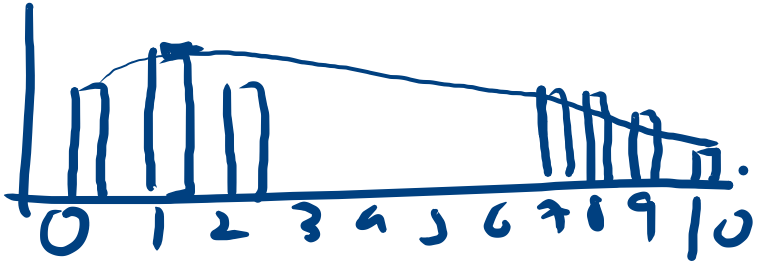
①

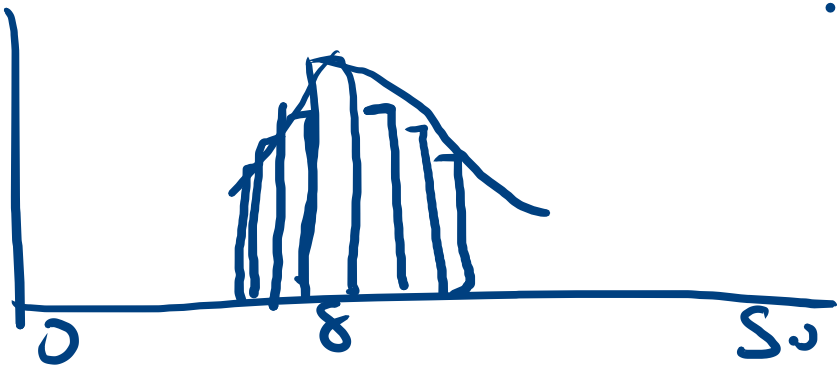


②



⑩



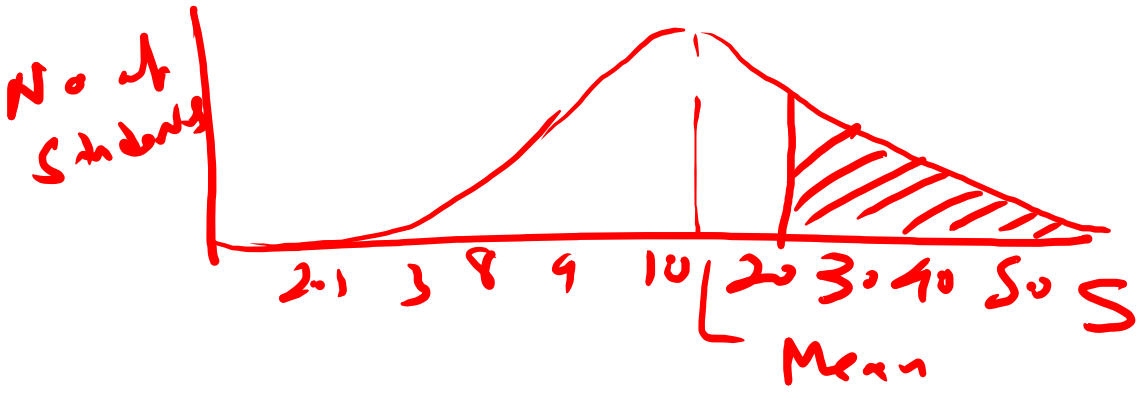


Lecture - 16

As n increases, the distribution of the sample mean $\bar{x}$ approaches a normal distribution.

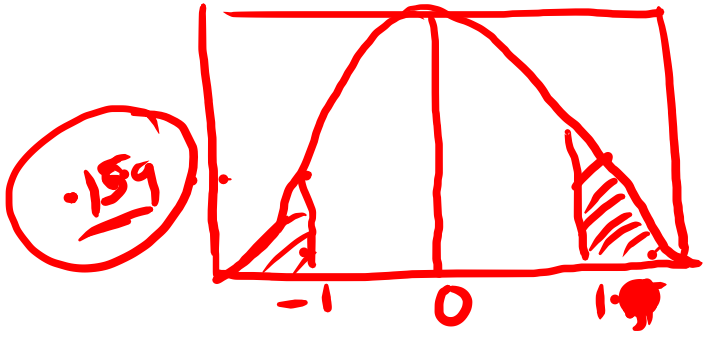
Central limit theorem.

Mean
Salary
= 10
SD
= 2 Lakh



RV

Two Students



Standard Normal

$N(0, 1)$

↓ Mean ↓ Var

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Probability of salary value less than (-1)

$$\int_{-\infty}^{-1} PDF$$

1.00 ← Total

Joint distribution &

1) $0 \le p_{x,y}(x_i, y_j) \le 1$

2) $\sum_{i,j} p_{x,y}(x_i, y_j) = 1$

3) $\sum_{j} p_{x,y}(x_i, y_j) = p_x(x_i)$

$\sum_{i} p_{x,y}(x_i, y_j) = p_y(y_i)$

$p_x(x_i)$

Marginal PMF

Last Example (DRV)

4) $p_{X,Y}(x_i, y_j) = P(\{X = x_i\} \cap \{Y = y_j\})$

A B

$= p_X(x_i)\, p_Y(y_j)$

are independent if product of marginal

$\Leftrightarrow$ PMF = joint PMF

Convert the point 10 taken to the corresponding point in standard normal distribution

$$Z = \frac{X - \mu}{\sigma}$$

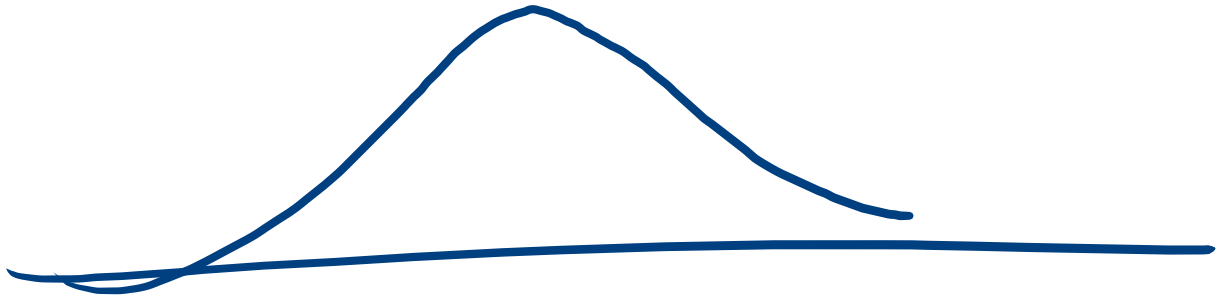
$X = 20$

$$= \frac{20 - 10}{4} = 2.5$$

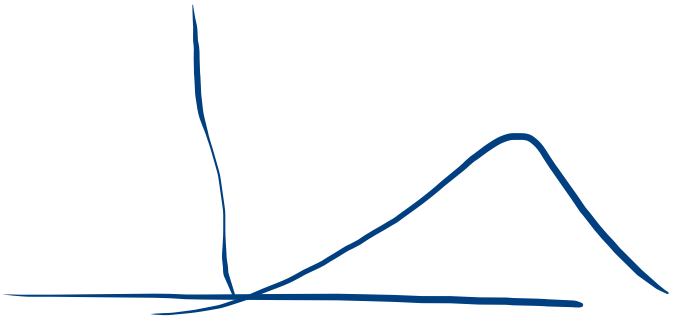
This is called Z Score.

Look into the table.

Lecture - 17





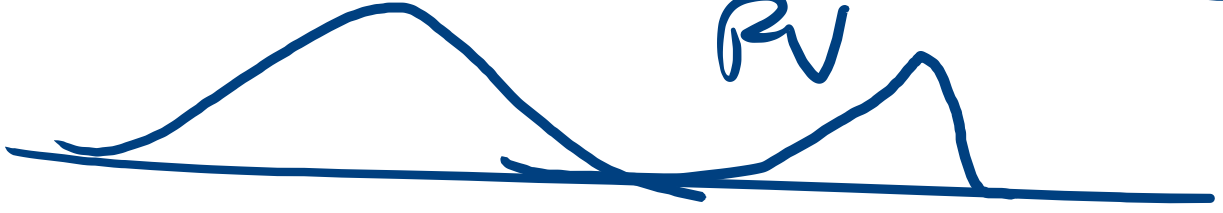


1) Location
2) Spread
3) Shape

Moments of PDF



RV



$E[x]$ is the best prediction of outcome of an experiment.

nth moment of a function $f(x)$ about a nonrandom value c

Raw Moments

$$E[(X-c)^n] = \sum_{\text{all } x_i} (x_i - c)^n p_X(x_i) \qquad \text{DRV}$$

$$= \int_{-\infty}^{\infty} (x_i - c)^n f_X(x_i)\, dx \qquad \text{CRV}$$

Support $c = 0$

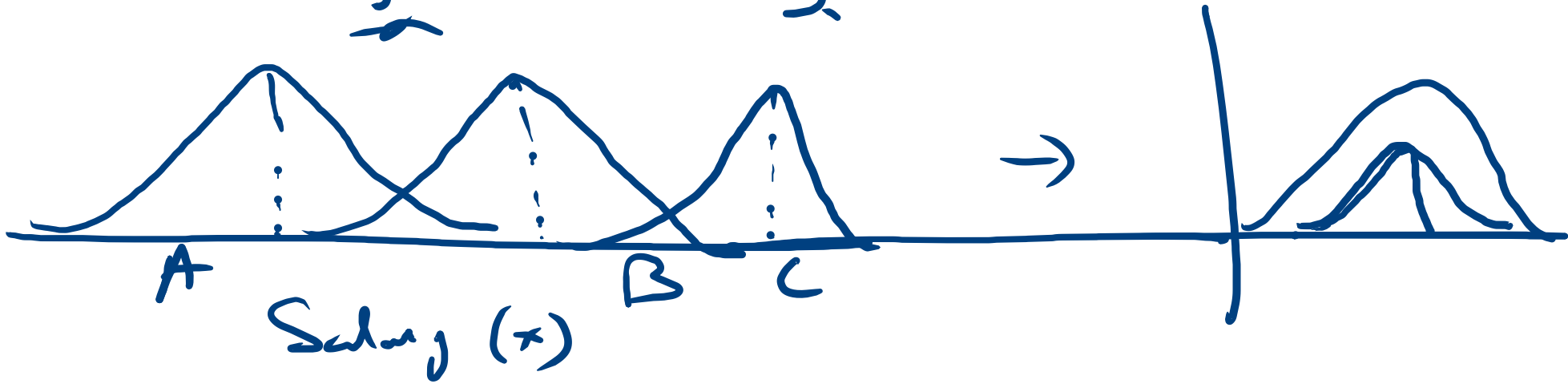
$$E[X^n] = \sum_{\text{all } x_i} x_i^n p_X(x_i) \\ = \int x_i^n f_X(x_i)\, dx$$

$\Rightarrow$ nth moment about $c = 0$ (origin)

1st moment about the origin — first Moment

$$E[X^1] = \sum_{i=1}^{\infty} x_i f_X(x_i) = \mu_X \quad \text{(Mean)}$$

$$= \int_{-\infty}^{\infty} x f_X(x) dx$$



* Scenario — when we want to compare the Variability of Salary between Ppl A & B

Central moment are useful because they allow us to quantify properties of distribution in ways that are location invariant.

$$E[(X-\mu)^n]$$

$E[(x-\mu)^2] \rightarrow$ Variance $\rightarrow \sigma_x^2$

$\sqrt{E[(x-\mu)^2]} \rightarrow$ SD $\rightarrow \sigma_x$

$$\boxed{\sigma_x^2 = E[x^2] - (E[x])^2}$$

Joint Distributions & Marginals

1) $0 \leq p_{X,Y}(x,y) \leq 1$

2) $\sum p_{X,Y}(x_i, y_j) = 1$

3) $\sum_y p_{X,Y}(x, y_j) = p_X(x)$

$\sum_x p_{X,Y}(x_i, y) = p_Y(y)$

} Marginal PMF

4) $p_{X,Y}(x_i, y_j) = p_X(x_i)\, p_Y(y_j)$

product of Marginals = Joint PMF

} X, Y are independent

Continuous RV (X, Y)

$$P(x-\Delta x < X < x+\Delta x,\ y-\Delta y < Y < y+\Delta y)$$

$$= \int_{x-\Delta x}^{x+\Delta x} \int_{y-\Delta y}^{y+\Delta y} f_{X,Y}(x,y)\, dy\, dx$$

$f_{X,Y}$ — Joint PDF

$$\int_{-\infty}^{\infty} f_{x,y}(x,y)\,dy = f_x(x) \quad \longrightarrow \text{Marg. n.r.}$$

Joint CDF

$$F_{x,y}(x,y) = P(X \le x,\ Y \le y)$$

properties of joint CDF $\rightarrow$ Assignment

Expected value of joint PDF

$$E[g(x)] = \sum_{\alpha} g(x) p_x(x) \qquad \text{DRV}$$

$$= \int_{-\infty}^{\infty} g(x) f_x(x) dx \qquad \text{CRV}$$

$g(x, y)$ where x & y are RVs

$$E[g(x,y)] = \sum_i \sum_j g(x_i, y_j) p_{x,y}(x_i, y_j)$$

↳ joint PMF

$$g(X,Y) = X+Y$$

$$E_{X,Y}[g(X,Y)] = E_{XY}[X+Y]$$

$$= \sum_i \sum_j (x_i + y_j)\, p_{XY}(x_i, y_j)$$

$$= \sum_i \sum_j x_i\, p_{XY}(x_i, y_j) + \sum_i \sum_j y_j\, p_{XY}(x_i, y_j)$$

$$= \sum_i x_i \sum_j p_{XY}(x_i, y_j) + \sum_j y_j \sum_i p_{XY}(x_i, y_j)$$

$$= \sum x_i\, p_X(x_i) + \sum y_j\, p_Y(y_j)$$

$$= E[X] + E[Y]$$

EV of Sum of RVs = Sum of EV of RVs

$$g(x,y) = xy$$

$$E[g(x,y)] = E[xy] = \sum_i \sum_j x_i y_j \, p_{xy}(x_i, y_j)$$

↳ **Correlation**

Can expected value of product of RVs
= product of EVs ?

$$E[g(x)\,h(y)] = E[g(x)]\,E[h(y)]$$

Variance of Sum of RV

$Z = X + Y$

$$Var(Z) = E[Z^2] - (E[Z])^2$$ ⟸ previous slide

$$= E[(X+Y)^2] - (E[X+Y])^2$$

$$= \cdots$$

$$= Var[X] + Var[Y] + 2\underbrace{[E[XY] - E[X]E[Y]]}_{\text{Covariance of } X,Y}$$

$$\boxed{Cov(X,Y) = E[XY] - E[X]E[Y]}$$

$Cov(X,Y)$

Joint moment of (X, Y)

1) about the origin
2) " " Mean

$$E[X^m Y^n] = \sum \sum x_i^m y_j^n \, p_{XY}(x_i, y_j)$$

$m=1, n=0 \Rightarrow E[X]$

$m=0, n=1 \Rightarrow E[Y]$

$m=1, n=1 \Rightarrow E[XY] \Rightarrow$ Correlation

$E[XY] = 0$, we say RVs are orthogonal.

about the mean

$$E\left[(X - E[X])^m (Y - E[Y])^n\right]$$

$m = 1, n = 0$ $\quad = E[X - E[X]] = 0$

$m = 1, n = 1$

Simplify it ....

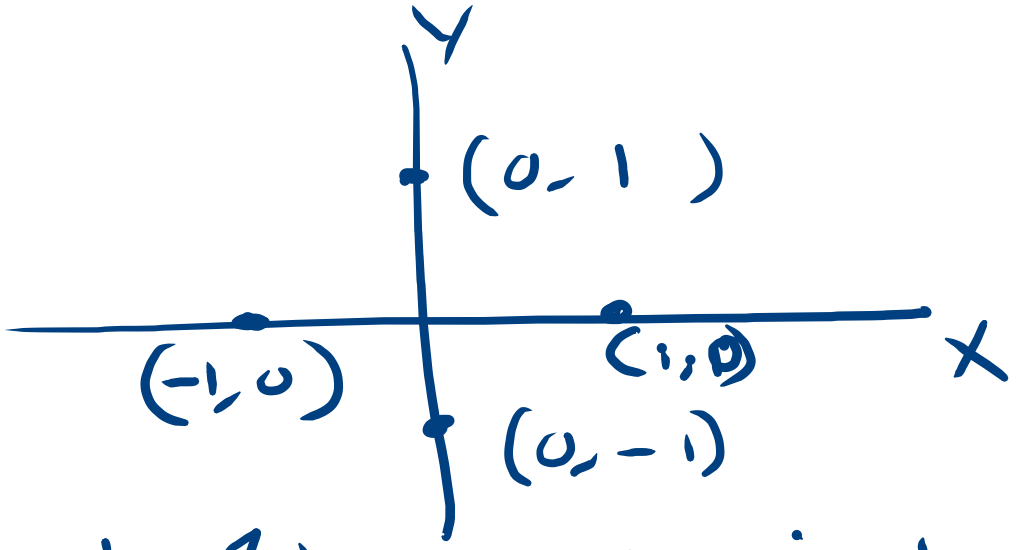
$$= E[XY] - E[X]\,E[Y]$$

covariance

If $Cov(X, Y)$ is Zero, we say X, Y uncorrelated

If X, Y are independent RVs, they are uncorrelated

however uncorrelated RVs does not imply independence



x-y plane

The moment I know $Y=1$, I know $X=0$

only instance $\Rightarrow$ they are dependent

Joint PMF

	$Y=-1$	$Y=0$	$Y=1$
$X=1$	0	$1/4$	0
$X=0$	$1/4$	0	$1/4$
$X=1$	0	$1/4$	0

x	-1	0	1	
$P_X(x)$	$1/4$	$1/2$	$1/4$	

y	-1	0	1	
$P_Y(y)$	$1/4$	$1/2$	$1/4$	

$$Cov(X,Y) = E[XY] - E[X]\,E[Y]$$

$$E[XY] = \sum_x \sum_y x_i y_j \, p_{XY}(x_i, y_j)$$

$$= \sum_{-1}^{1} \sum_{-1}^{1} - - p_{XY}(x_i, y_j)$$

$$= (-1)(-1)\,0 + (-1)(0)\left(\frac{1}{4}\right) + \cdots \text{9 terms}$$

You will find a 0 in each term

$$= 0 - 0 = 0$$

Lecture - 18

$\Rightarrow$ $Cov(X,Y) = E[XY] - E[X]\,E[Y]$

↳ Correlation ↳ Avg value of X,Y

Correlation is a similarity measure.

* $Cov(X,Y)$ contains the correlation term, $Cov = 0$ does not imply independence.

$$Cov(X,Y) = E\left[(X - E(X))(Y - E(Y))\right]$$

$\rho_{X,Y}$ = correlation coefficient

$$= E\left[\frac{(X - E(X))}{\sigma_X}\left(Y - \frac{E(Y)}{\sigma_Y}\right)\right]$$

$$-1 \leq \rho_{x,y} \leq 1$$